E-commerce Purchase Prediction using Machine Learning

In [1]:
```
import pandas as pd
import numpy as np

from sklearn.model_selection import train_test_split
from sklearn.compose import ColumnTransformer
from sklearn.preprocessing import StandardScaler, OneHotEncoder
from sklearn.pipeline import Pipeline
from sklearn.tree import DecisionTreeClassifier
from sklearn.metrics import f1_score, classification_report, confusion_matrix
```

In [2]:
```
df = pd.read_csv("shop_smart_ecommerce.csv")
```

In [3]:
```
df.head()
```

Out[3]:

	Administrative	Administrative_Duration	Informational	Informational_Duration	Prod
0	0	0.0	0	0.0	
1	0	0.0	0	0.0	
2	0	0.0	0	0.0	
3	0	0.0	0	0.0	
4	0	0.0	0	0.0	

In [4]:
```
df.tail(5)
```

Out[4]:

	Administrative	Administrative_Duration	Informational	Informational_Duration
12325	3	145.0	0	0.0
12326	0	0.0	0	0.0
12327	0	0.0	0	0.0
12328	4	75.0	0	0.0
12329	0	0.0	0	0.0

In [5]:
```
df.shape
```

Out[5]:
```
(12330, 18)
```

In [6]:
```
df.describe()
```

Out[6]:

	Administrative	Administrative_Duration	Informational	Informational_Duration
count	12330.000000	12330.000000	12330.000000	12330.000000
mean	2.315166	80.818611	0.503569	34.472398
std	3.321784	176.779107	1.270156	140.749294
min	0.000000	0.000000	0.000000	0.000000
25%	0.000000	0.000000	0.000000	0.000000
50%	1.000000	7.500000	0.000000	0.000000
75%	4.000000	93.256250	0.000000	0.000000
max	27.000000	3398.750000	24.000000	2549.375000

In [7]:
```
df.info()
```

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 12330 entries, 0 to 12329
Data columns (total 18 columns):
 #   Column                   Non-Null Count  Dtype
---  ------                   --------------  -----
 0   Administrative           12330 non-null  int64
 1   Administrative_Duration  12330 non-null  float64
 2   Informational            12330 non-null  int64
 3   Informational_Duration   12330 non-null  float64
 4   ProductRelated           12330 non-null  int64
 5   ProductRelated_Duration  12330 non-null  float64
 6   BounceRates              12330 non-null  float64
 7   ExitRates                12330 non-null  float64
 8   PageValues               12330 non-null  float64
 9   SpecialDay               12330 non-null  float64
 10  Month                    12330 non-null  object
 11  OperatingSystems         12330 non-null  int64
 12  Browser                  12330 non-null  int64
 13  Region                   12330 non-null  int64
 14  TrafficType              12330 non-null  int64
 15  VisitorType              12330 non-null  object
 16  Weekend                  12330 non-null  bool
 17  Revenue                  12330 non-null  bool
dtypes: bool(2), float64(7), int64(7), object(2)
memory usage: 1.5+ MB
```

In [8]:
```
df.isna().sum()
```

Out[8]:
```
Administrative             0
Administrative_Duration    0
Informational              0
Informational_Duration     0
ProductRelated             0
ProductRelated_Duration    0
BounceRates                0
ExitRates                  0
PageValues                 0
SpecialDay                 0
Month                      0
OperatingSystems           0
Browser                    0
Region                     0
TrafficType                0
VisitorType                0
Weekend                    0
Revenue                    0
dtype: int64
```

In [9]:
```
df.columns
```

Out[9]:
```
Index(['Administrative', 'Administrative_Duration', 'Informational',
       'Informational_Duration', 'ProductRelated', 'ProductRelated_Duration',
       'BounceRates', 'ExitRates', 'PageValues', 'SpecialDay', 'Month',
       'OperatingSystems', 'Browser', 'Region', 'TrafficType', 'VisitorType',
       'Weekend', 'Revenue'],
      dtype='object')
```

In [10]:
```
X = df.drop(columns=["Revenue"])
y = df["Revenue"].astype(int)
```

In [11]:
```
num_features = X.select_dtypes(include=["int64", "float64"]).columns
cat_features = X.select_dtypes(include=["object", "category"]).columns
```

In [12]:
```
X_train, X_test, y_train, y_test = train_test_split(
    X,
    y,
    test_size=0.2,
    random_state=42,
    stratify=y
)
```

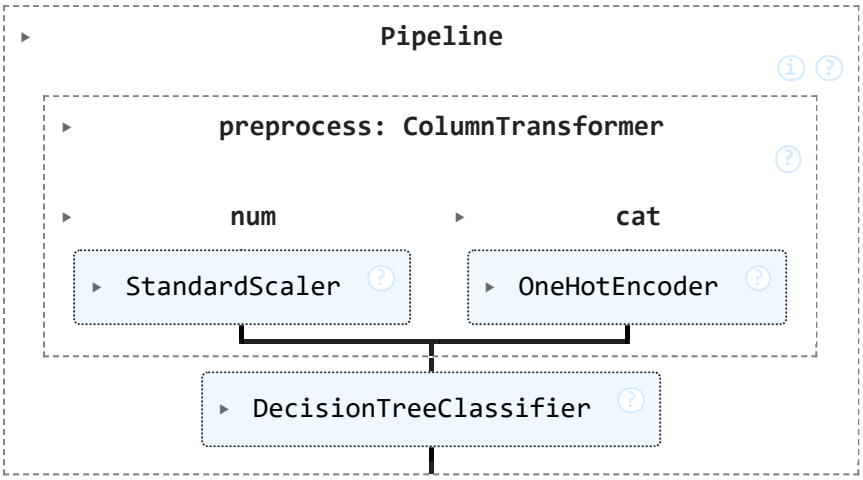
In [13]:
```
# Preprocessng Pipeline
preprocessor = ColumnTransformer(
    transformers=[
        ("num", StandardScaler(), num_features),
        ("cat", OneHotEncoder(handle_unknown="ignore"), cat_features)
    ]
)
```

In [14]:
```
dt = DecisionTreeClassifier(
    max_depth=6,              # prevents deep overfitting
    min_samples_leaf=30,      # smooths decision boundaries
    class_weight="balanced",  # handles imbalance
    random_state=42
)
```

In [15]:
```
pipe = Pipeline(
    steps=[
        ("preprocess", preprocessor),
        ("model", dt)
    ]
)
```

In [16]:
```
pipe.fit(X_train, y_train)
```

Out[16]:



In [17]:
```
y_pred = pipe.predict(X_test)

print("F1 score:", f1_score(y_test, y_pred))
print("\nClassification Report:\n", classification_report(y_test, y_pred))
print("\nConfusion Matrix:\n", confusion_matrix(y_test, y_pred))
```

```
F1 score: 0.6278381046396841

Classification Report:
               precision    recall  f1-score   support

           0       0.97      0.85      0.90      2084
           1       0.50      0.83      0.63       382

    accuracy                           0.85      2466
   macro avg       0.73      0.84      0.77      2466
weighted avg       0.89      0.85      0.86      2466


Confusion Matrix:
 [[1771  313]
 [  64  318]]
```

Hyperparameter Tuning

In [18]:
```
from sklearn.model_selection import GridSearchCV

param_grid = {
    "model__max_depth": [4, 6, 8],
    "model__min_samples_leaf": [20, 30, 50]
}
```

In [19]:
```
grid = GridSearchCV(
    pipe,
```

```
    param_grid,
    scoring="f1",
    cv=5,
    n_jobs=-1
)

grid.fit(X_train, y_train)

print("Best F1:", grid.best_score_)
print("Best params:", grid.best_params_)
```

```
Best F1: 0.6343735129725652
Best params: {'model__max_depth': 4, 'model__min_samples_leaf': 50}
```

In [ ]: